

Computational Astrophysics Exercises

Assignments Set 6 — Summer Term 2026

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Date: 19.06.26

Exercise 1: The Restricted Circular Three Body Problem

10 pts

(2a.)

We implemented an RK4 integrator for the RC3BP equations of motion with $\mu_2 = 10^{-3}$, $C_J = 3.03$, $\Delta t = 10^{-3}$, and initial conditions $\dot{x}_0 = 0$, $y_0 = 0$. The following figures show particle positions sampled at every 1st, 10th, and 500th step, with increasing total step counts.

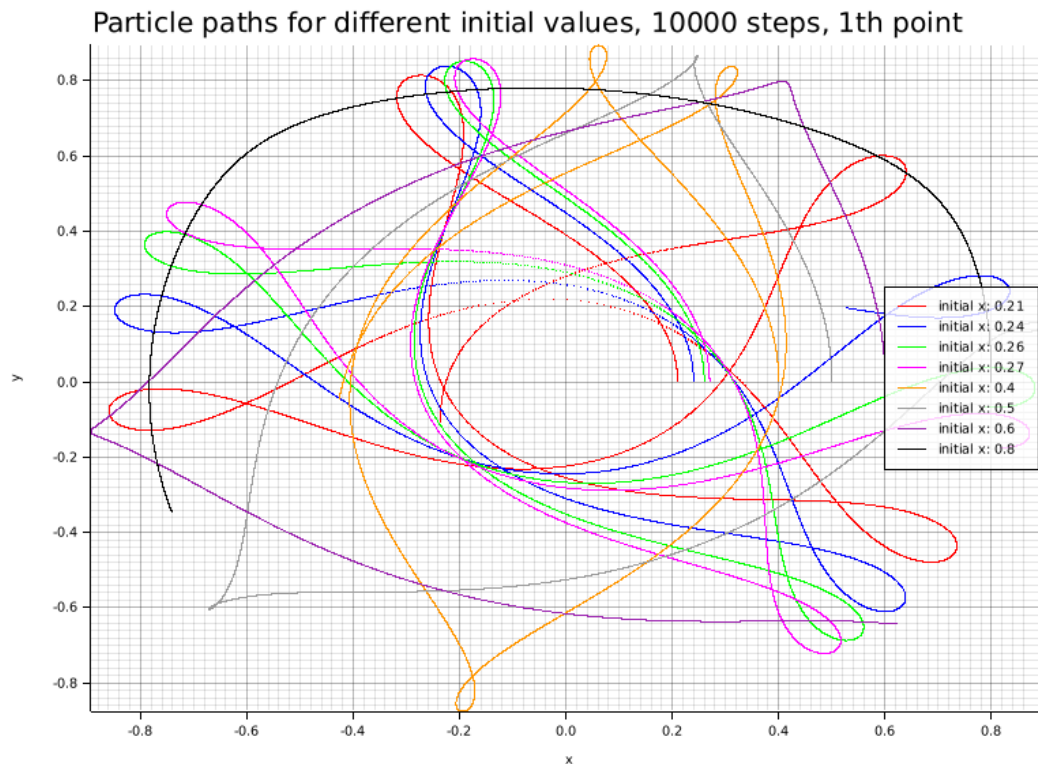


Figure 1: Particle paths for 10,000 steps with every point plotted. Trajectories are still short, the structure of individual orbits is visible.

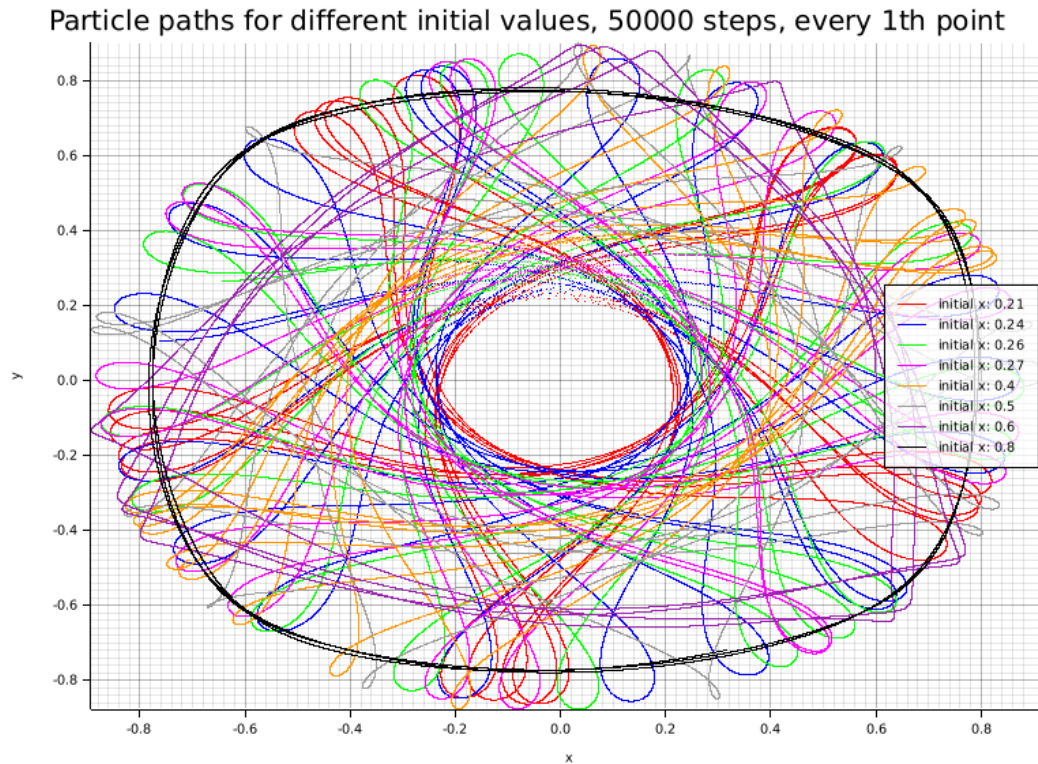


Figure 2: 50,000 steps, every point. Orbits begin to fill out their allowed regions. An hollow inner exclusion zone and outer boundary become apparent.

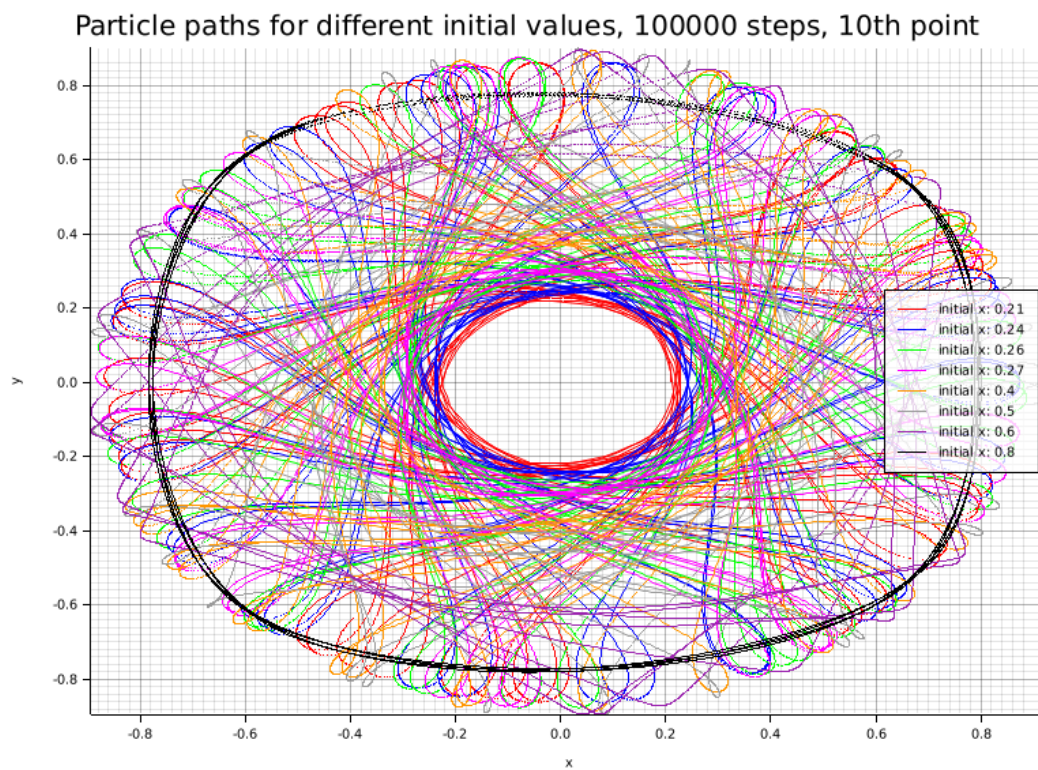


Figure 3: 100,000 steps, every 10th point. The torus-like structure becomes clear. Most initial conditions show bounded curves, indicating periodicity.

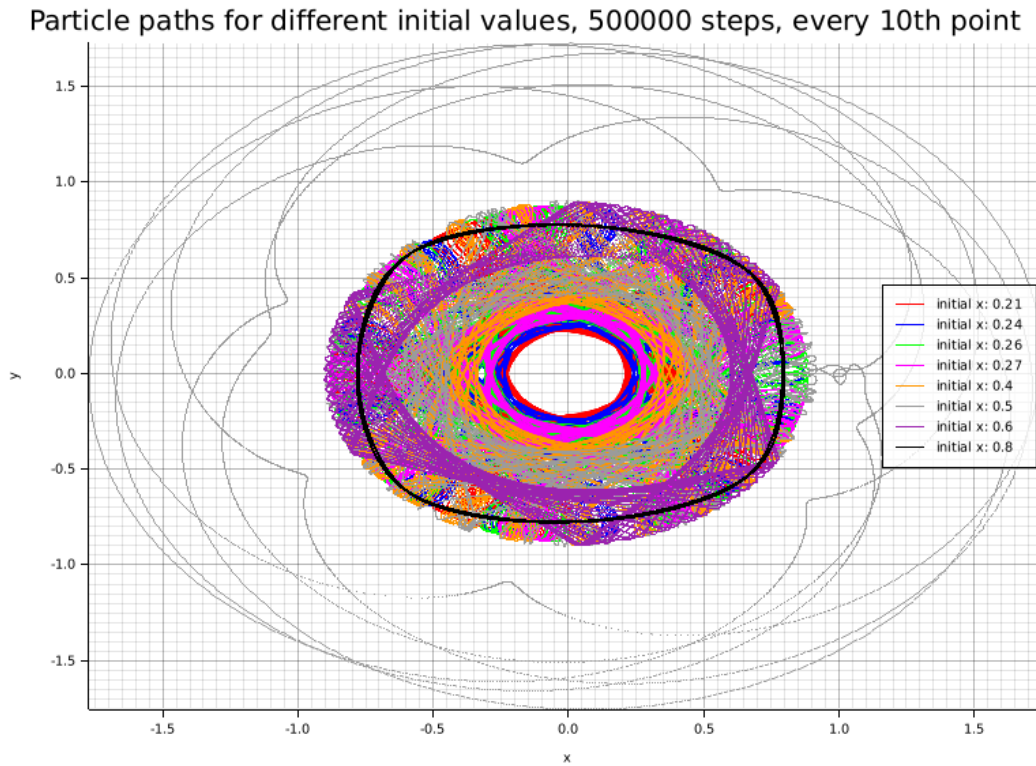


Figure 4: 500,000 steps, every 10th point. The grey trajectory ($x_0 = 0.5$) has escaped to large radii. All other initial conditions remain bounded.

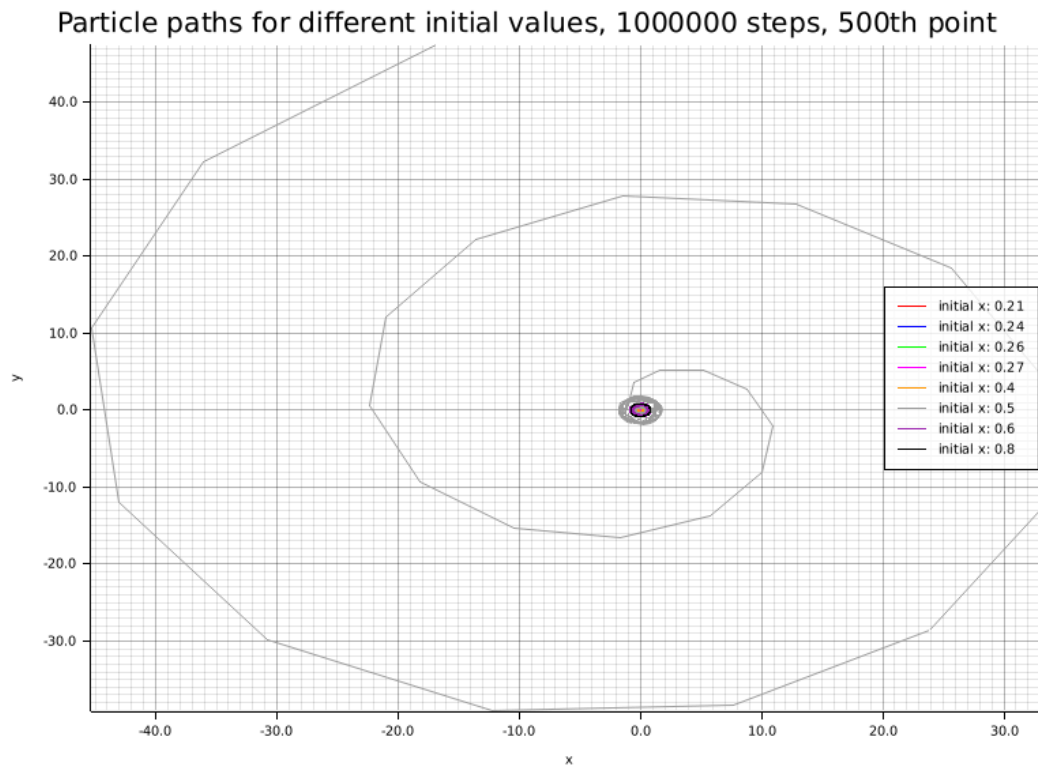


Figure 5: 1,000,000 steps, every 500th point. The escaping grey orbit now completely diverges.

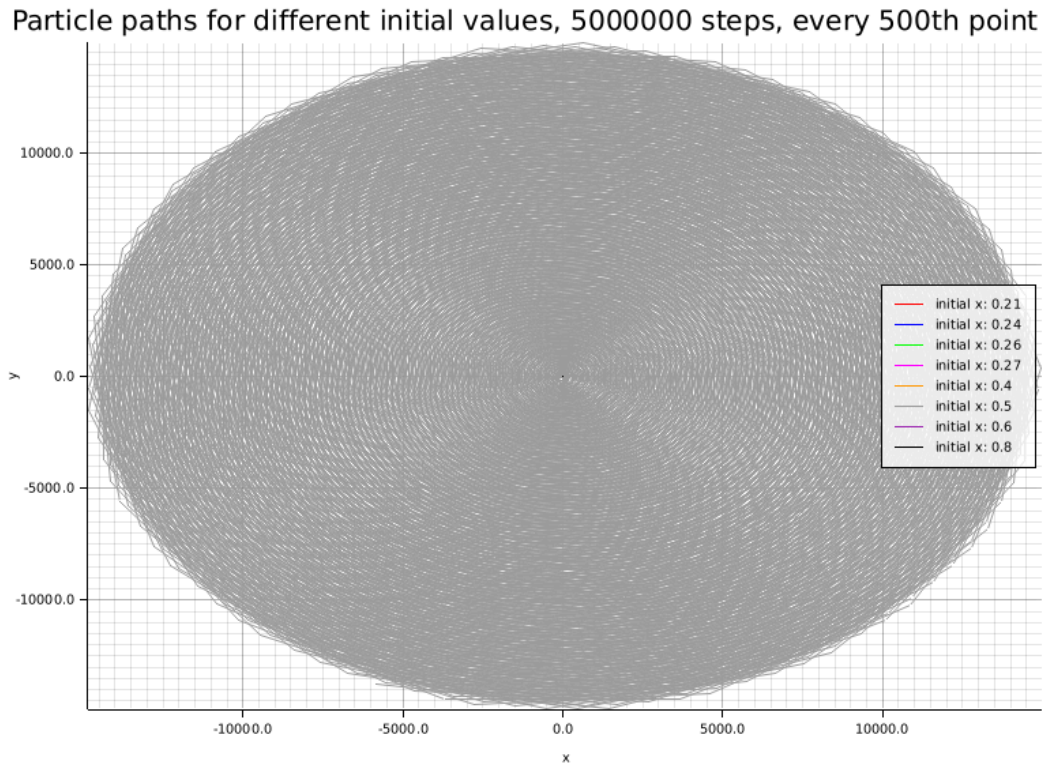


Figure 6: 5,000,000 steps, every 500th point. The grey ($x_0 = 0.5$) orbit has completely escaped, confirming it is chaotic and unbound on long timescales.

Symmetry: The trajectories show a reflection symmetry about the x -axis. This follows from the equations of motion: The y -equation is odd in y , so if $(x(t), y(t))$ is a solution with $\dot{y}_0 > 0$, $(-x(t), y(t))$ with $\dot{y}_0 < 0$ is also a solution.

Stability: Initial conditions $x_0 \in \{0.21, 0.24, 0.26, 0.27, 0.4, 0.6, 0.8\}$ yield bounded, periodic trajectories. The initial condition $x_0 = 0.5$ produces an orbit that escapes to large distances for large step counts, thus indicating chaotic or unstable behaviour.

(2b.)

Poincaré sections were constructed by recording $(x, \dot{x})$ each time the particle crosses $y = 0$ with $\dot{y} > 0$.

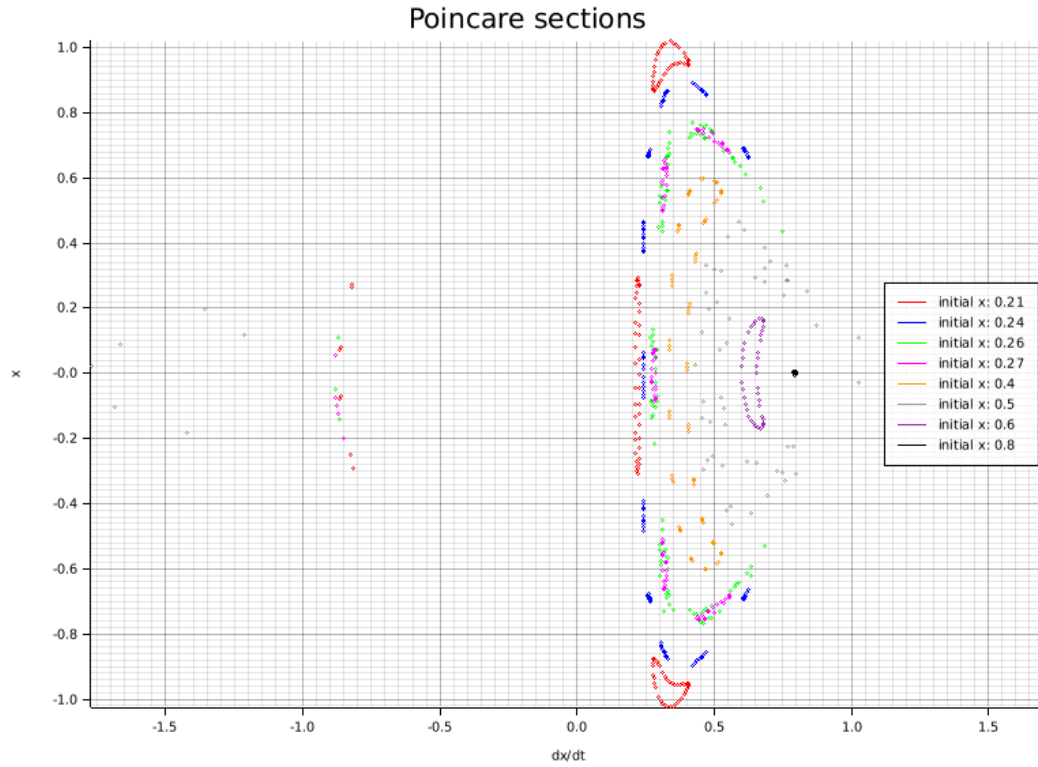


Figure 7: Poincaré section for steps until $5 \cdot 10^5$. Stable orbits trace closed invariant curves around $\dot{x} \approx 0.3$ - 0.5 . Loops can be seen for *red*, *blue*, *green*, *purple*, *yellow*, *pink* and *black* is concentrated into a point. The *gray* ($x_0 = 0.5$) section is chaotic, this becomes even more apparent in Figure 8

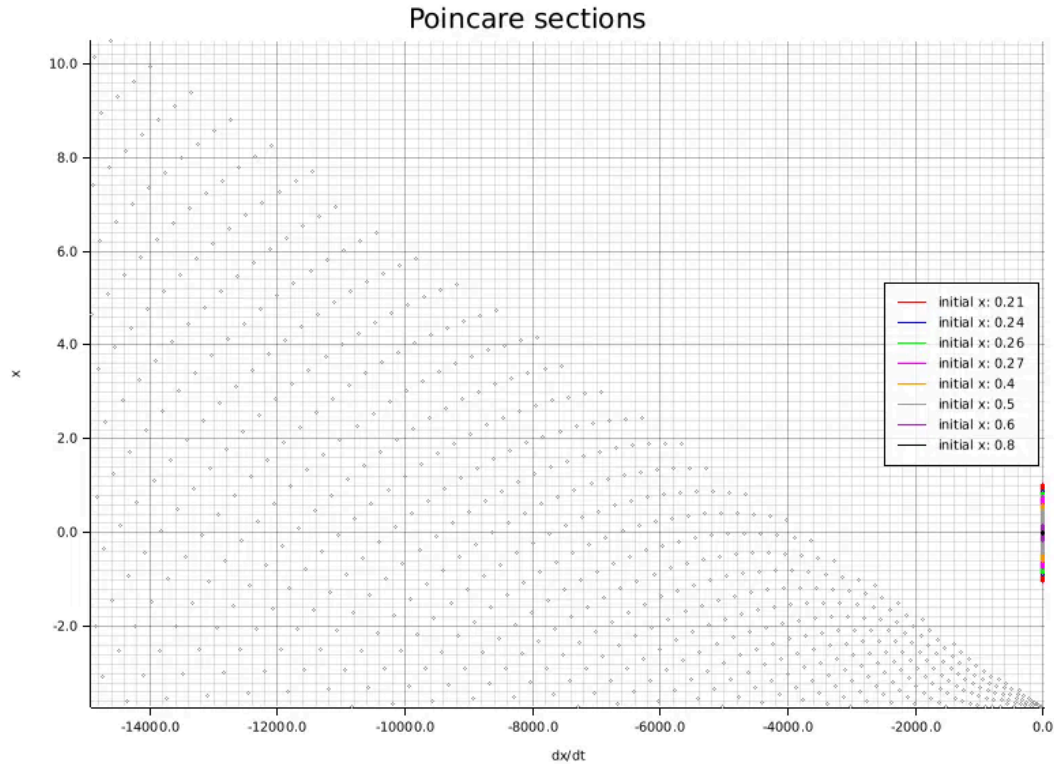


Figure 8: Poincaré section for larger step counts, until $5 \cdot 10^6$. This clearly shows the chaos of the initial value $x_0 = 0.5$

This analysis shows the stability more clearly:

- **Stable orbits** are $x_0 \in \{0.21, 0.24, 0.26, 0.27, 0.4, 0.6, 0.8\}$, as the crossings lie on closed curves in the range $\dot{x} \approx 0.2$ - 0.6 . The looping curves (resonant islands) can be seen and traced, although with gaps in between. The clearest islands are *red* ($x_0 = 0.21$) and *purple* ($x_0 = 0.6$).
- **The unstable orbit** of $x_0 = 0.27$ is scattered broadly across the plane, first bounded in Figure 7 and then completely divergent in Figure 8.

(Extra)

We implemented a small visualizer for the three bodies. This shows the two large masses in fixed position, while the small mass orbits around them.